Homework 4

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Question 1

Consider an experiment with two factors, each with two levels.

Consider the two-factor linear model with interaction

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon, \qquad \varepsilon \sim \text{i.i.d. } (0, \sigma^2).$$

Compare two designs with coded levels in $[-1, 1]$:

- **Design A (factorial):** full 2^2 factorial with coded levels $\{-1, +1\}$ for each factor, with r replicates per cell (total $n = 4r$).
- **Design B (random):** $n = 4r$ runs where $x_1, x_2 \sim \text{Unif}[-1, 1]$, independent across runs.

Simulation part. Fix $(\beta_0, \beta_1, \beta_2, \beta_{12})$ and σ^2 (provide your chosen values), and $r \in \{1, 2, 5\}$. For each r:

- Generate 5,000 datasets from Design A and from Design B.
- Fit OLS including the interaction.

For the simulation, I chose the following coefficient values and error variance:

$$(\beta_0, \beta_1, \beta_2, \beta_{12}) = (10, 2, -1, 1.5) \qquad \sigma^2 = 1$$

Here, I provide the R source code used for the simulation:

R code

```r
## Question 1: simulation only
set.seed(2026)

## 1) Choose a truth and simulation settings
beta <- c(10, 2, -1, 1.5)
names(beta) <- c("(Intercept)", "x1", "x2", "x1:x2")
sigma <- 1
r_vals <- c(1, 2, 5)
n_sim <- 5000

## 2) Design A table (coded levels)
desA <- expand.grid(x1 = c(-1, 1), x2 = c(-1, 1))
X_A <- model.matrix(~ x1 * x2, desA)

## 3) Run the simulation and store raw coefficient estimates
sim_out <- vector("list", length(r_vals))
names(sim_out) <- paste0("r=", r_vals)

for (j in seq_along(r_vals)) {
  r <- r_vals[j]
  N <- 4 * r
  beta_hat_A <- matrix(NA_real_, nrow = n_sim, ncol = length(beta))
  beta_hat_B <- matrix(NA_real_, nrow = n_sim, ncol = length(beta))
  colnames(beta_hat_A) <- names(beta)
  colnames(beta_hat_B) <- names(beta)

  for (i in seq_len(n_sim)) {
    ## Design A: replicated 2^2 factorial
    X_rep <- X_A[rep(seq_len(nrow(X_A)), each = r), ]
    datA <- data.frame(
      x1 = X_rep[, "x1"],
      x2 = X_rep[, "x2"]
    )
    datA$Y <- as.vector(X_rep %*% beta) + rnorm(N, 0, sigma)
    beta_hat_A[i, ] <- coef(lm(Y ~ x1 * x2, data = datA))

    ## Design B: random runs on [-1, 1]^2
    datB <- data.frame(
      x1 = runif(N, min = -1, max = 1),
      x2 = runif(N, min = -1, max = 1)
    )
    X_B <- model.matrix(~ x1 * x2, datB)
    datB$Y <- as.vector(X_B %*% beta) + rnorm(N, 0, sigma)
    beta_hat_B[i, ] <- coef(lm(Y ~ x1 * x2, data = datB))
  }

  sim_out[[j]] <- list(
    r = r,
    factorial = as.data.frame(beta_hat_A, check.names = FALSE),
    random = as.data.frame(beta_hat_B, check.names = FALSE)
  )
}
## 4) Peek at the simulated coefficient estimates
cat("\nFirst 5 coefficient estimates for Design A (Factorial), r = 1\n")
print(head(sim_out[["r=1"]]$factorial, 5))
cat("\nFirst 5 coefficient estimates for Design B (Random), r = 1\n")
print(head(sim_out[["r=1"]]$random, 5))
```

Here is the output:

R output

```
First 5 coefficient estimates for Design A (Factorial), r = 1
  (Intercept)       x1         x2     x1:x2
1    9.873847 1.543933 -0.8466022 1.8440732
2    9.588217 2.972376 -1.1881567 2.4745657
3   10.936345 2.046770 -1.0418001 0.6168311
4    9.863515 1.278142 -0.9873413 1.5207784
5    9.962437 2.147705 -1.6504864 1.8341441

First 5 coefficient estimates for Design B (Random), r = 1
  (Intercept)         x1         x2       x1:x2
1   10.143883  -1.180347 -0.3777064  -3.2709492
2   10.451018   4.230248 -9.8206288  27.3408651
3   10.631182   1.213685 -0.6206000   3.3269354
4   18.164955 -17.983303 21.3745644 -10.1747675
5    9.571763   2.945335 -1.6920467  -0.1957562
```

a. Report empirical bias and variance for each coefficient under each design.

For computing *empirical bias* and *empirical variance* I will use the following R code:

R code

```r
## 5) q1.a: empirical bias and variance for each coefficient under each
↪  design
summarize_coef <- function(est_df, true_beta, design_name, r) {
  est_mat <- as.matrix(est_df)
  data.frame(
    r = r,
    design = design_name,
    coefficient = colnames(est_mat),
    bias = colMeans(est_mat) - true_beta[colnames(est_mat)],
    variance = apply(est_mat, 2, var),
    row.names = NULL,
    check.names = FALSE
  )
}
q1a_out <- do.call(
  rbind,
  lapply(
    sim_out,
    function(obj) {
      rbind(
        summarize_coef(obj$factorial, beta, "Factorial", obj$r),
        summarize_coef(obj$random, beta, "Random", obj$r)
      )
    }
  )
)
print(q1a_out)
```

Here the output:

R output

```
       r    design coefficient          bias     variance
r=1.1 1 Factorial (Intercept) -1.815053e-03 2.500017e-01
r=1.2 1 Factorial          x1  7.757159e-03 2.544645e-01
r=1.3 1 Factorial          x2 -8.644509e-03 2.482329e-01
r=1.4 1 Factorial       x1:x2 -1.511875e-03 2.498113e-01
r=1.5 1    Random (Intercept) -1.903427e+01 1.188840e+06
r=1.6 1    Random          x1  3.765812e+01 1.421082e+07
r=1.7 1    Random          x2 -2.497530e+01 2.011497e+06
r=1.8 1    Random       x1:x2  9.685221e+01 2.648259e+07
r=2.1 2 Factorial (Intercept)  1.011710e-02 1.268268e-01
r=2.2 2 Factorial          x1  9.396195e-03 1.205481e-01
r=2.3 2 Factorial          x2 -1.076806e-03 1.322803e-01
r=2.4 2 Factorial       x1:x2 -1.295276e-04 1.242788e-01
r=2.5 2    Random (Intercept)  1.001492e-03 4.309794e-01
r=2.6 2    Random          x1 -2.588629e-02 1.608886e+00
r=2.7 2    Random          x2  9.686902e-03 1.571549e+00
r=2.8 2    Random       x1:x2 -8.626820e-03 6.390041e+00
r=5.1 5 Factorial (Intercept) -4.554305e-03 4.916849e-02
r=5.2 5 Factorial          x1 -3.911694e-04 4.980880e-02
r=5.3 5 Factorial          x2  2.687862e-03 4.769173e-02
r=5.4 5 Factorial       x1:x2 -5.621014e-03 4.957038e-02
r=5.5 5    Random (Intercept)  1.371518e-03 6.113999e-02
r=5.6 5    Random          x1 -1.047613e-03 1.987865e-01
r=5.7 5    Random          x2 -6.636974e-03 2.007188e-01
r=5.8 5    Random       x1:x2  3.520744e-03 6.866970e-01
```

b. Explain why bias is (approximately) zero for both designs and why variance is smaller under the factorial design.

The empirical bias is approximately zero for both designs because the fitted OLS model matches the true mean model used to generate the data, and the error term has mean zero. Therefore, under the correctly specified model, the OLS estimators are unbiased.

The variance is smaller under the factorial design because the coded levels produce an orthogonal design. This orthogonality makes the columns of the design matrix uncorrelated, so $(X^\top X)^{-1}$ is more stable, which reduces the variance of the coefficient estimates. In contrast, the random design is not guaranteed to be orthogonal, so the columns of the design matrix can be correlated, making $(X^\top X)^{-1}$ less stable and increasing the variance of the estimates. The very large values for the random design when $r = 1$ occur because there are only $n = 4$ observations and four fitted parameters, so the fit is saturated and numerically unstable.

Question 2

An article in *Industrial and Engineering Chemistry* ("More on Planning Experiments to Increase Research Efficiency," 1970, pp. 60–65) uses a 2^{5-2} design to investigate the effect of A = condensation temperature, B = amount of material 1, C = solvent volume, D = condensation time, and E = amount of material 2 on yield. The results obtained are as follows:

$$\begin{array}{cccc} e = 23.2 & ad = 16.9 & cd = 23.8 & bde = 16.8 \\ ab = 15.5 & bc = 16.2 & ace = 23.4 & abcde = 18.1 \end{array}$$

a. Verify that the design generators used were $I = ACE$ and $I = BDE$.

To verify this, we construct the coded design table and compute the columns ACE and BDE:

A	B	C	D	E	ACE	BDE
−1	−1	−1	−1	+1	+1	+1
+1	−1	−1	+1	−1	+1	+1
−1	−1	+1	+1	−1	+1	+1
−1	+1	−1	+1	+1	+1	+1
+1	+1	−1	−1	−1	+1	+1
−1	+1	+1	−1	−1	+1	+1
+1	−1	+1	−1	+1	+1	+1
+1	+1	+1	+1	+1	+1	+1

Adding the run labels makes it clear that the observed design corresponds to these generators.

Run	A	B	C	D	E	ACE	BDE
e	−1	−1	−1	−1	+1	+1	+1
ad	+1	−1	−1	+1	−1	+1	+1
cd	−1	−1	+1	+1	−1	+1	+1
bde	−1	+1	−1	+1	+1	+1	+1
ab	+1	+1	−1	−1	−1	+1	+1
bc	−1	+1	+1	−1	−1	+1	+1
ace	+1	−1	+1	−1	+1	+1	+1
abcde	+1	+1	+1	+1	+1	+1	+1

Since both computed columns are +1 for every run, the defining generators are indeed $I = ACE$ and $I = BDE$.

b. Write down the complete defining relation and the aliases for this design.

Starting from the defining words $I = ACE$ and $I = BDE$, we obtain

$$I = ACE = BDE = ABCD$$

Multiplying the defining relation by each main effect gives the alias classes for the main effects:

$$\begin{aligned}
A &= CE = ABDE = BCD \\
B &= ABCE = DE = ACD \\
C &= AE = BCDE = ABD \\
D &= ACDE = BE = ABC \\
E &= AC = BD = ABCDE
\end{aligned}$$

The remaining alias classes are obtained in the same way:

$$\begin{aligned}
AB &= CD = ADE = BCE \\
AD &= BC = ABE = CDE
\end{aligned}$$

These equations give the complete alias structure of the design.

c. Estimate the main effects.

For each factor, the main effect is the average response at the high level minus the average response at the low level. With eight runs, this is the corresponding contrast divided by 4.

$$\begin{aligned}
A &= \frac{ad + ab + ace + abcde - (e + cd + bde + bc)}{4} \\
A &= \frac{16.9 + 15.5 + 23.4 + 18.1 - (23.2 + 23.8 + 16.8 + 16.2)}{4} \\
A &= \frac{-6.1}{4} \\
A &= -1.525
\end{aligned}$$

$$\begin{aligned}
B &= \frac{bde + ab + bc + abcde - (e + ad + cd + ace)}{4} \\
B &= -5.175
\end{aligned}$$

$$\begin{aligned}
C &= \frac{cd + bc + ace + abcde - (e + ad + bde + ab)}{4} \\
C &= 2.275
\end{aligned}$$

$$D = \frac{ad + cd + bde + abcde - (e + ab + bc + ace)}{4}$$
$$D = -0.675$$

$$E = \frac{e + bde + ace + abcde - (ad + cd + ab + bc)}{4}$$
$$E = 2.275$$

If the coded regression model is fit with levels $\{-1, +1\}$, the corresponding regression coefficients are one-half of these main-effect estimates.

Question 3

Consider the following experiment:

Run	**Treatment Combination**
1	d
2	ae
3	b
4	$abde$
5	cde
6	ac
7	bce
8	$abcd$

Answer the following questions about this experiment:

a. How many factors does this experiment investigate?

This experiment investigates five factors: A, B, C, D, and E. This follows from the five distinct letters that appear in the treatment combinations.

b. How many independent factors are in this design?

An independent factor is a factor whose levels are assigned freely in the design. Since this design has 8 runs and is a two-level fractional factorial, it has $\log_2 8 = 3$ independent factors.

c. What are the generators for the dependent factors?

Taking A, B, and C as the independent factors, we can treat D and E as dependent factors:

Run	A	B	C	D	E	AB	AC
d	-1	-1	-1	$+1$	-1	$+1$	$+1$
ae	$+1$	-1	-1	-1	$+1$	-1	-1
b	-1	$+1$	-1	-1	-1	-1	$+1$
$abde$	$+1$	$+1$	-1	$+1$	$+1$	$+1$	-1
cde	-1	-1	$+1$	$+1$	$+1$	$+1$	-1
ac	$+1$	-1	$+1$	-1	-1	-1	$+1$
bce	-1	$+1$	$+1$	-1	$+1$	-1	-1
$abcd$	$+1$	$+1$	$+1$	$+1$	-1	$+1$	$+1$

As shown in the table, $D = AB$ and $E = -AC$. These are the generators.

d. Is this design a principal fraction?

No. The all-low treatment (1) is not included, so this is not the principal fraction.

e. What is the complete defining relation and aliases?

Combining $D = AB$ and $E = -AC$, we obtain

$$I = ABD = -ACE = -BCDE$$

The alias classes for the main effects are therefore

$$\begin{aligned}
I &= ABD = -ACE = -BCDE \\
A &= BD = -CE = -ABCDE \\
B &= AD = -CDE = -ABCE \\
C &= -AE = -BDE = ABCD \\
D &= AB = -BCE = -ACDE \\
E &= -AC = -BCD = ABDE
\end{aligned}$$

The remaining alias classes are

$$\begin{aligned}
BC &= -DE = -ABE = ACD \\
BE &= -CD = -ABC = ADE
\end{aligned}$$

f. What is the resolution?

The resolution is III because the shortest word in the defining relation $I = ABD = -ACE = -BCDE$ has length 3; specifically, ABD and ACE each contain three letters.

So we can conclude that this is a 2^{5-2}_{III} fractional factorial design.

Question 4

Construct a 2^{7-2} design by choosing two four-factor interactions as the independent generators. Write down the complete alias structure for this design. What is the resolution of this design?

To construct the design, choose the two generators

$$A = BCDE$$
$$F = BCDG$$

Now generate the design table from these generators, noting that A and F are dependent factors and that B, C, D, E, and G are independent factors.

Run	A	B	C	D	E	F	G
1	+1	−1	−1	−1	−1	+1	−1
2	−1	+1	−1	−1	−1	−1	−1
3	−1	−1	+1	−1	−1	−1	−1
4	+1	+1	+1	−1	−1	+1	−1
5	−1	−1	−1	+1	−1	−1	−1
6	+1	+1	−1	+1	−1	+1	−1
7	+1	−1	+1	+1	−1	+1	−1
8	−1	+1	+1	+1	−1	−1	−1
9	−1	−1	−1	−1	+1	+1	−1
10	+1	+1	−1	−1	+1	−1	−1
11	+1	−1	+1	−1	+1	−1	−1
12	−1	+1	+1	−1	+1	+1	−1
13	+1	−1	−1	+1	+1	−1	−1
14	−1	+1	−1	+1	+1	+1	−1
15	−1	−1	+1	+1	+1	+1	−1
16	+1	+1	+1	+1	+1	−1	−1
17	+1	−1	−1	−1	−1	−1	+1
18	−1	+1	−1	−1	−1	+1	+1
19	−1	−1	+1	−1	−1	+1	+1
20	+1	+1	+1	−1	−1	−1	+1
21	−1	−1	−1	+1	−1	+1	+1
22	+1	+1	−1	+1	−1	−1	+1
23	+1	−1	+1	+1	−1	−1	+1
24	−1	+1	+1	+1	−1	+1	+1
25	−1	−1	−1	−1	+1	−1	+1
26	+1	+1	−1	−1	+1	+1	+1
27	+1	−1	+1	−1	+1	+1	+1
28	−1	+1	+1	−1	+1	−1	+1
29	+1	−1	−1	+1	+1	+1	+1
30	−1	+1	−1	+1	+1	−1	+1
31	−1	−1	+1	+1	+1	−1	+1
32	+1	+1	+1	+1	+1	+1	+1

From the generators $A = BCDE$ and $F = BCDG$, the corresponding defining words are

$$I = AEFG = ABCDE = BCDFG$$

Multiplying each effect by the defining relation gives the alias classes for the main effects:

$$\begin{aligned}
A &= EFG = BCDE = ABCDFG \\
B &= ACDE = CDFG = ABEFG \\
C &= ABDE = BDFG = ACEFG \\
D &= ABCE = BCFG = ADEFG \\
E &= AFG = ABCD = BCDEFG \\
F &= AEG = BCDG = ABCDEF \\
G &= AEF = BCDF = ABCDEG
\end{aligned}$$

The alias classes for the two-factor interactions are

$$\begin{aligned}
AB &= CDE = BEFG = ACDFG \\
AC &= BDE = CEFG = ABDFG \\
AD &= BCE = DEFG = ABCFG \\
AE &= FG = BCD = ABCDEFG \\
AF &= EG = ABCDG = BCDEF \\
AG &= EF = ABCDF = BCDEG \\
BC &= ADE = DFG = ABCEFG \\
BD &= ACE = CFG = ABDEFG \\
BE &= ACD = ABFG = CDEFG \\
BF &= CDG = ABEG = ACDEF \\
BG &= CDF = ABEF = ACDEG \\
CD &= ABE = BFG = ACDEFG \\
CE &= ABD = ACFG = BDEFG \\
CF &= BDG = ACEG = ABDEF \\
CG &= BDF = ACEF = ABDEG \\
DE &= ABC = ADFG = BCEFG \\
DF &= BCG = ADEG = ABCEF \\
DG &= BCF = ADEF = ABCEG
\end{aligned}$$

The remaining alias classes are

$$
\begin{aligned}
ABF &= BEG = ACDG = CDEF \\
ABG &= BEF = ACDF = CDEG \\
ACF &= CEG = ABDG = BDEF \\
ACG &= CEF = ABDF = BDEG \\
ADF &= DEG = ABCG = BCEF \\
ADG &= DEF = ABCF = BCEG
\end{aligned}
$$

The design has resolution IV because the shortest word in the defining relation $I = AEFG = ABCDE = BCDFG$ has length 4.

Therefore, this is a 2^{7-2}_{IV} fractional factorial design.